

AgriSmart AI – Project Planning & Management

Project: AgriSmart AI – Intelligent Crop Recommendation System **Document Type:** Project Planning & Management Documentation **Submitted for:** DEPI Project Documentation Requirements **Team Size:** 6 members + supervisor **Duration:** 8 weeks

1. Project Proposal

1.1 Project Overview

AgriSmart AI is an AI-based crop recommendation system. It uses soil and environmental parameters, such as nitrogen, phosphorus, potassium, pH, temperature, humidity, and rainfall, to recommend the most suitable crop for given field conditions. The system will be built with machine learning, will explain its predictions using Explainable AI techniques, and will be delivered through a simple web application so that non-technical users can obtain recommendations easily.

1.2 Problem Statement

Selecting a suitable crop depends on the interaction of many soil and environmental factors. These factors vary between locations and seasons, and their combined effect is difficult to judge manually. Decisions based on habit, limited experience, or incomplete information can lead to poor crop suitability, inefficient use of land and inputs, and reduced productivity. A data-driven tool that analyzes these parameters together can support more informed and consistent crop selection.

1.3 Project Goal

To develop an intelligent system that recommends suitable crops from soil and environmental parameters using machine learning.

1.4 Objectives

No.	Objective
O1	Analyze soil and environmental data to understand the factors that influence crop suitability.

No.	Objective
O2	Develop and compare multiple machine learning models for crop recommendation.
O3	Select and optimize the most appropriate model using systematic evaluation and tuning.
O4	Provide explainable predictions so that users can understand why a crop was recommended.
O5	Develop a simple, user-friendly application that delivers recommendations.

1.5 Project Scope

In Scope	Out of Scope
Data collection and understanding	Crop disease detection
Exploratory Data Analysis (EDA)	Yield prediction
Data preprocessing	Fertilizer recommendation
Machine learning model development	Market price prediction
Model evaluation	Farmer marketplace
Hyperparameter tuning	Weather forecasting as a standalone system
Explainable AI (XAI)	
Application development	
Testing	
Documentation	

2. Project Plan

The project is planned as an 8-week schedule, starting from Week 1 (Day 1) of the project. Each week has a defined focus, a set of activities, concrete deliverables, and a responsible team member.

2.1 Weekly Plan

Week	Focus	Activities	Deliverables	Responsible Team Member
1	Project Planning & Data Understanding	Define scope and objectives; prepare plan and roles; set up GitHub repository; collect the dataset; review features, data types, and target classes	Project Proposal; Project Plan; dataset description	Member 1 (data); all members (planning)
2	EDA & Data Visualization	Analyze distributions, correlations, and class balance; detect outliers and missing values; create visualizations	EDA Report; visualization set	Member 2
3	Data Preprocessing & Feature Engineering	Clean data; handle outliers; scale and encode features; perform feature selection or engineering; split into train and test sets	Preprocessing pipeline; processed dataset	Member 3
4	Baseline Machine Learning Models	Train baseline classifiers; establish initial performance benchmarks	Baseline models; baseline results table	Member 4
5	Model Comparison & Hyperparameter Tuning	Compare candidate models (including XGBoost); apply cross-validation and tuning; select the best model	Comparison report; tuned best model	Member 4

Week	Focus	Activities	Deliverables	Responsible Team Member
6	Model Evaluation & Explainable AI	Evaluate with accuracy, precision, recall, F1-score, and confusion matrix; apply SHAP to explain predictions	Evaluation report; XAI analysis	Member 5
7	Application Development & Integration	Build the Streamlit interface; integrate the trained model and XAI output; handle input validation	Working Streamlit application	Member 6
8	Testing, Documentation & Final Presentation	Test functionality and predictions; fix defects; finalize documentation; prepare and deliver presentation	Testing Report; final documentation; Final Presentation	All members (Member 6 leads testing)

2.2 Gantt-Style Schedule

■ indicates planned active work in that week.

Activity	W1	W2	W3	W4	W5	W6	W7	W8
Project planning	■							
Data collection & understanding	■	■						
EDA & visualization		■	■					
Preprocessing & feature engineering			■	■				
Baseline ML models				■	■			
Model comparison & tuning					■	■		
Model evaluation						■		
Explainable AI						■	■	
Application development & integration							■	■
Testing							■	■
Documentation	■	■	■	■	■	■	■	■
Final presentation								■

3. Milestones

ID	Milestone	Target Week	Completion Criterion
M1	Project Planning Completed	Week 1	Proposal, plan, and roles approved by the supervisor
M2	Dataset Understanding Completed	Week 1-2	Dataset described and features documented
M3	EDA Completed	Week 2-3	EDA report and visualizations delivered
M4	Preprocessing Completed	Week 3-4	Reproducible preprocessing pipeline and train/test split ready
M5	ML Models Developed	Week 4-5	Baseline and candidate models trained and recorded
M6	Best Model Selected	Week 5-6	Tuned model chosen based on evaluation results
M7	Explainable AI Integrated	Week 6-7	SHAP explanations produced and linked to the model

ID	Milestone	Target Week	Completion Criterion
M8	Application Completed	Week 7–8	Streamlit app runs end-to-end with the selected model
M9	Testing & Documentation Completed	Week 8	Testing report and final documentation submitted

4. Deliverables

Deliverable	Description	Planned Completion
Project Proposal	Overview, problem statement, goal, objectives, and scope	Week 1
EDA Report	Data analysis findings, statistics, and visualizations	Week 2–3
Preprocessing	Cleaned dataset and documented preprocessing pipeline	Week 3–4
ML Models	Baseline and tuned machine learning models with comparison results	Week 4–5
Model Evaluation	Performance metrics, confusion matrix, and model selection justification	Week 6
XAI	SHAP-based explanation of model predictions and key features	Week 6–7
Streamlit Application	Web interface that accepts inputs and returns recommendations with explanations	Week 7–8
Testing Report	Test cases, results, identified issues, and fixes	Week 8
Documentation	Planning, technical, and user documentation	Week 8
Final Presentation	Summary presentation of the project and its outcomes	Week 8

5. Team Roles & Task Assignment

Member	Role	Main Responsibilities	Primary Deliverables
Member 1	Data Collection & Data Understanding	Source the dataset; verify quality; document features, ranges, and target classes	Dataset description; data dictionary
Member 2	EDA & Visualization	Perform exploratory analysis; identify patterns, outliers, and class balance; produce visualizations	EDA Report
Member 3	Preprocessing & Feature Engineering	Clean and transform data; scale features; select or engineer features; split data	Preprocessing pipeline; processed dataset
Member 4	Machine Learning & Hyperparameter Tuning	Train and compare models; perform cross-validation and tuning	ML models; comparison results; tuned model
Member 5	Model Evaluation & Explainable AI	Evaluate models with appropriate metrics; apply SHAP; interpret results	Evaluation report; XAI analysis
Member 6	Streamlit Application & System Integration	Design and build the app; integrate model and XAI outputs; lead testing	Streamlit application; Testing Report

All members contribute to project planning in Week 1 and to documentation and the final presentation in Week 8. The supervisor provides guidance and reviews milestones.

6. Resource Allocation

Category	Resources	Purpose
Human Resources	6 project team members and 1 supervisor	Execution, guidance, and review

Category	Resources	Purpose
Software	Python, Jupyter Notebook, VS Code, GitHub, Streamlit	Development, experimentation, version control, and application delivery
Libraries	Pandas, NumPy, Scikit-learn, XGBoost, SHAP, Matplotlib, Seaborn	Data handling, modeling, explainability, and visualization
Hardware	Standard development computers or laptops and an internet connection	Development and collaboration

The project requires no specialized or costly infrastructure. All listed software and libraries are freely available.

7. Risk Assessment & Mitigation

Risk	Probability	Impact	Mitigation
Poor data quality	Medium	High	Validate data early in Weeks 1-2; document issues; apply cleaning and outlier handling in preprocessing
Low model performance	Medium	High	Compare several algorithms; apply feature engineering and hyperparameter tuning; review results weekly
Overfitting	Medium	High	Use a held-out test set and cross-validation; compare training and validation scores
Integration problems	Medium	Medium	Agree on model and input formats early; save the model in a standard format; test integration incrementally
API/internet issues	Low	Medium	Keep the dataset and code stored locally; avoid dependence on external services at runtime; keep offline backups

Risk	Probability	Impact	Mitigation
Time management	Medium	High	Track tasks weekly; prioritize core features; begin documentation from Week 1
Team coordination problems	Medium	Medium	Hold weekly meetings; assign clear owners; use GitHub for shared work and communication

8. Key Performance Indicators (KPIs)

KPI	Description	Target
Model Accuracy / F1 Score	Predictive quality of the selected model on unseen data	Achieve strong, consistent accuracy and F1-score across all crop classes, without signs of overfitting
Prediction Response Time	Time between user input and displayed recommendation	Return recommendations quickly enough for smooth interactive use
Application Availability	Ability of the application to run reliably when launched	Application runs without critical failures during demonstration and testing
Successful Test Cases	Proportion of planned test cases that pass	Pass all critical test cases; resolve identified defects before final submission
Documentation Completion	Coverage of required documentation	All required documents completed and reviewed by the end of Week 8
Milestone Completion Rate	Proportion of milestones completed on schedule	Complete all nine milestones within the planned timeline

9. Project Monitoring

Mechanism	Description
Weekly team meetings	Review progress, discuss obstacles, and plan the next week's tasks
GitHub version control	Maintain code, notebooks, and documents in a shared repository with tracked changes
Weekly progress review	Compare completed work with the weekly plan and report status to the supervisor
Task assignment	Assign each task to a named member with a clear deadline
Milestone tracking	Check each milestone against its completion criterion and record the date achieved
Risk review	Reassess the risk table weekly and update mitigation actions as needed

10. Conclusion

This project plan organizes the AgriSmart AI team by assigning each of the six members a clear role aligned with a stage of the machine learning workflow. The 8-week schedule, milestones, and deliverables provide a structured path from data understanding to the final application. The risk assessment identifies the main threats early and defines practical mitigation actions, while the KPIs and monitoring mechanisms allow progress and quality to be tracked consistently. Together, these elements support coordinated work and the successful delivery of a documented, explainable, and user-friendly crop recommendation system.